

Ch.7 Zero-Sum Games

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Zero-Sum Games

Minimax and Maximin

Mixed Strategies

Minimax Theorem

An Example

- ▶ Two players: I and II
- ▶ The sum of their payoffs is always 0.
- ▶ In the payoff matrix, only I's payoff is shown.
- ▶ It's a strictly competitive game.
- ▶ von Neumann: Does the value exist?

Security Level

$$M = \begin{array}{c|ccc} & t_1 & t_2 & t_3 \\ \hline s_1 & 1 & 6 & 0 \\ s_2 & 2 & 0 & 3 \\ s_3 & 3 & 2 & 4 \end{array}$$

- ▶ Def: A player's security level is the largest expected payoff he can guarantee to himself whatever the other player may do.
- ▶ For player I,
 - ▶ $s_1: \min\{1, 6, 0\} = 0$
 - ▶ $s_2: \min\{2, 0, 3\} = 0$
 - ▶ $s_3: \min\{3, 2, 4\} = 2$
 - ▶ I's security level $= \max\{0, 0, 2\} = 2$
- ▶ $S \equiv \{s_1, s_2, s_3\}$, $T \equiv \{t_1, t_2, t_3\}$
- ▶ $\pi(s, t)$: I's payoff for the strategy profile (s, t)
- ▶ I's security level $= \max_{s \in S} \min_{t \in T} \pi(s, t) \equiv \underline{m}$
- ▶ $\underline{m}$ is also called the maximin value of matrix M

Security Level of II

▶

		t_1	t_2	t_3
$M =$	s_1	1	6	0
	s_2	2	0	3
	s_3	3	2	4

- ▶ For player II,
- ▶ $t_1: \max\{1, 2, 3\} = 3$
 - ▶ $t_2: \max\{6, 0, 2\} = 6$
 - ▶ $t_3: \max\{0, 3, 4\} = 4$
 - ▶ II's security level = $\min\{3, 6, 4\} = 3$
- ▶ II's security level = $\min_{t \in T} \max_{s \in S} \pi(s, t) \equiv \overline{m}$
- ▶ $\overline{m}$ is also called the minimax value of matrix M

Minimax and Maximin

- ▶ In this example, $\overline{m} = 3 > 2 = \underline{m}$.
- ▶ In general, $\overline{m} \geq \underline{m}$
- ▶ Proof: Consider $\underline{m}$ is in the i -th row, and $\overline{m}$ is in the j -th column of matrix M .
Consider the element m_{ij} of M .
We have $\underline{m} \leq m_{ij} \leq \overline{m}$. $\square$
- ▶ What happens if $\underline{m} = \overline{m}$?

- ▶ Let (σ, τ) denote a strategy profile and let $\pi(\sigma, \tau)$ denote the payoff to I.
- ▶ Def: (σ, τ) is a saddle point of M if

$$\pi(\sigma, \tau) = \max_{s \in S} \pi(s, \tau) = \min_{t \in T} \pi(\sigma, t)$$

- ▶ Theorem: (a) $\underline{m} = \overline{m} \Leftrightarrow \exists$ saddle point
(b) If (σ, τ) is a saddle point, then $\pi(\sigma, \tau) = \underline{m} = \overline{m}$

Proof: (a) Suppose $\underline{m} = \overline{m}$.

Let σ be I's security strategy, i.e. $\min_{t \in T} \pi(\sigma, t) = \underline{m}$.

Let τ be II's security strategy, i.e. $\max_{s \in S} \pi(s, \tau) = \overline{m}$.

$$\Rightarrow \overline{m} \geq \pi(\sigma, \tau) \geq \underline{m}$$

$$\Rightarrow \overline{m} = \pi(\sigma, \tau) = \underline{m}, \text{ since } \underline{m} = \overline{m}$$

$$\Rightarrow \pi(\sigma, \tau) = \max_{s \in S} \pi(s, \tau) = \min_{t \in T} \pi(\sigma, t), (\sigma, \tau) \text{ is a saddle point.}$$

On the other hand, if (σ, τ) is a saddle point,

$$\pi(\sigma, \tau) = \min_{t \in T} \pi(\sigma, t) \leq \underline{m}$$

$$\pi(\sigma, \tau) = \max_{s \in S} \pi(s, \tau) \geq \overline{m}$$

So, $\underline{m} \geq \overline{m}$. But we know that $\underline{m} \leq \overline{m}$, so $\underline{m} = \overline{m}$.

(b) If (σ, τ) is a saddle point, from above,

$$\overline{m} \leq \pi(\sigma, \tau) \leq \underline{m}.$$

Since $\underline{m} = \overline{m}$ from (a), $\pi(\sigma, \tau) = \underline{m} = \overline{m}$. $\square$

- ▶ In our example, $\overline{m} > \underline{m}$, so saddle point does not exist.
- ▶ Von Neumann pressed to seek a saddle point in the product space of mixed strategies.
- ▶ Def: $P \equiv \{(p_1, p_2, p_3) : \sum p_i = 1, p_i \geq 0\}$ as I's strategy space.
- ▶ Def: $Q \equiv \{(q_1, q_2, q_3) : \sum q_i = 1, q_i \geq 0\}$ as II's strategy space.
- ▶ Def: Payoff function $\pi : P * Q \rightarrow R$
- ▶ $\pi(p, q) = pMq^T$
- ▶

$$\text{Ex. } M = \begin{pmatrix} 1 & 6 & 0 \\ 2 & 0 & 3 \\ 3 & 2 & 4 \end{pmatrix}, \quad p = (1/4, 1/4, 1/2), \quad q = (1/2, 1/2, 0)$$

- ▶ With p , when II uses t_1 , t_2 , or t_3 , I's expected payoff is:

$$(1/4, 1/4, 1/2) \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \frac{9}{4}, \quad (1/4, 1/4, 1/2) \begin{pmatrix} 6 \\ 0 \\ 2 \end{pmatrix} = \frac{5}{2}, \quad (1/4, 1/4, 1/2) \begin{pmatrix} 0 \\ 3 \\ 4 \end{pmatrix} = \frac{11}{4}$$

- ▶ Consider q , I's expected payoff is:

$$\left(\frac{9}{4}, \frac{5}{2}, \frac{11}{4}\right) \begin{pmatrix} 1/2 \\ 1/2 \\ 0 \end{pmatrix} = 2\frac{3}{8}$$

Déjà vu

- ▶ Def: I's security level $\underline{v} \equiv \max_{p \in P} \min_{q \in Q} \pi(p, q)$
- ▶ Def: II's security level $\overline{v} \equiv \min_{q \in Q} \max_{p \in P} \pi(p, q)$
- ▶ Let $\tilde{p}$, $\tilde{q}$ be I's and II's security strategy.

$$\underline{v} = \min_{q \in Q} \pi(\tilde{p}, q), \quad \overline{v} = \max_{p \in P} \pi(p, \tilde{q})$$

- ▶ $\Rightarrow \underline{v} \leq \pi(\tilde{p}, \tilde{q}) \leq \overline{v}$

Déjà vu

- ▶ Def: $(\tilde{p}, \tilde{q})$ is a saddle point for $\pi(., .)$ if
$$\pi(\tilde{p}, q) \geq \pi(\tilde{p}, \tilde{q}) \geq \pi(p, \tilde{q}), \forall p \in P, \forall q \in Q.$$
- ▶ Theorem: (a) $\underline{v} = \overline{v} \Leftrightarrow \exists$ saddle point
(b) If $(\tilde{p}, \tilde{q})$ is a saddle point, then $\pi(\tilde{p}, \tilde{q}) = \underline{v} = \overline{v}$

► Minimax Theorem: $\underline{v} = \overline{v}$.

► Proof: When M is 1×1 , clearly $\underline{v} = \overline{v}$.

Mathematical induction: suppose the theorem holds for all submatrices M' of $M_{m \times n}$, we'll show that it then also holds for M .

Suppose not, we then have:

$$\underline{v} < \overline{v}.$$

Let $\tilde{p}$ and $\tilde{q}$ denote I's and II's security strategies:

$$\underline{v} \leq \pi(\tilde{p}, \tilde{q}) \leq \overline{v}.$$

These inequalities implies one of the following must be true:

$$\underline{v} < \pi(\tilde{p}, \tilde{q}), \quad \pi(\tilde{p}, \tilde{q}) < \overline{v}.$$

We'll assume the latter is true. The proof is similar if the former is true.

$$\begin{aligned} \pi(\tilde{p}, \tilde{q}) &= \tilde{p}(M\tilde{q}^T) \\ &= \tilde{p}_1 M_{1.} \tilde{q}^T + \dots + \tilde{p}_m M_{m.} \tilde{q}^T. \end{aligned}$$

$\pi(\tilde{p}, \tilde{q}) < \overline{v}$ implies $\exists M_{k.}$, s.t. $M_{k.} \tilde{q}^T < \overline{v}$.

Let $s = \overline{v} - M_{k.} \tilde{q}^T > 0$.

Let M' be M w/o M_k . Let p' and q' be the security strategies of M' . By assumption M' 's value v' exists.

Note p' is $1 * (m - 1)$. To proceed smoothly, we'll expand p' to $1 * m$ by inserting 0 to be its k -th element.

Consider $r \in (0, 1)$ and define:

$$\begin{aligned} p'' &= (1 - r)\tilde{p} + rp', \\ q'' &= (1 - r)\tilde{q} + rq'. \end{aligned}$$

$$\begin{aligned} \min_j p'' M_{.j} &\geq \min_j (1 - r)\tilde{p} M_{.j} + \min_j r p' M_{.j} \\ &= (1 - r)\underline{v} + rv'. \\ \max_{i \neq k} M_{i.} q''^T &\leq \max_{i \neq k} (1 - r) M_{i.} \tilde{q} + \max_{i \neq k} r M_{i.} q'^T \\ &\leq (1 - r)\overline{v} + rv'. \end{aligned} \tag{1}$$

Consider a very large number $a = \max_{ij} M_{ij} + 1$.

$$\begin{aligned} M_{k.} q''^T &< (1 - r)(\overline{v} - s) + ra \\ &= (1 - r)\overline{v} + rv' - (1 - r)s + r(a - v'). \end{aligned}$$

Note $-(1 - r)s + r(a - v') \uparrow$ in r , and it is 0 when $r = s/[(a - v') + s] \in (0, 1)$.

Consider $r \in (0, s/[(a - v') + s])$, then

$$\max_i M_{i.} q''^T \leq (1 - r)\overline{v} + rv'. \tag{2}$$

From (1) and (2),

$$\begin{aligned}\min_j p'' M_{.j} &\geq (1-r)\underline{v} + rv', \\ \max_i M_i q''^T &\leq (1-r)\bar{v} + rv'.\end{aligned}$$

Thus

$$\begin{aligned}\max_i M_i q''^T - \min_j p'' M_{.j} &\leq (1-r)(\bar{v} - \underline{v}) \\ &< \bar{v} - \underline{v}\end{aligned}$$

At least one of the following inequality must hold:

$$\max_i M_i q''^T < \bar{v} \text{ or } \min_j p'' M_{.j} > \underline{v}.$$

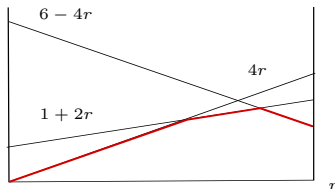
But, neither can be true. $\square$

- In the following example, we have shown that $\overline{m} = 3 > 2 = \underline{m}$.

		t_1	t_2	t_3
$M =$	s_1	1	6	0
	s_2	2	0	3
	s_3	3	2	4

- Minimax Thm. tells us that $\overline{v} = \underline{v}$. What is the value?
- What is I's mixed security strategy?
- I plays s_2 w.p.0.
 - I's security strategy is: $p = (1 - r, 0, r)$.
- Given r , let's imagine the worst situation for I.
- If II plays t_1 , $\pi = (1 - r) + 3r = 1 + 2r$.
 - t_2 , $\pi = 6(1 - r) + 2r = 6 - 4r$.
 - t_3 , $\pi = 0 + 4r = 4r$.
- Given r , I's worst expected payoff is $\min\{1 + 2r, 6 - 4r, 4r\}$.
- Do we have to consider II's mixed strategy?

- ▶ The red line in following graph shows $\min\{1 + 2r, 6 - 4r, 4r\}$ for different r .



- ▶ What is I's security strategy?
- ▶ $6 - 4r = 1 + 2r \Rightarrow r = 5/6$
- ▶ $\underline{v} = 8/3$

$$M = \begin{array}{c|ccc} & t_1 & t_2 & t_3 \\ \hline s_1 & 1 & 6 & 0 \\ s_2 & 2 & 0 & 3 \\ s_3 & 3 & 2 & 4 \end{array}$$

- ▶ From the Minimax Them., we know II's security level $\bar{v} = 8/3$.
- ▶ What is II's security strategy, $q = (q_1, q_2, q_3)$?
 - ▶ If I plays s_1 , $\pi = q_1 + 6q_2$.
 - ▶ s_3 , $\pi = 3q_1 + 2q_2 + 4q_3$
 - ▶ Do we have to consider s_2 ?
- ▶ Given q , the worst situation for II is that I receives $\max\{q_1 + 6q_2, 3q_1 + 2q_2 + 4q_3\}$, or $\max\{q_1 + 6q_2, 4 - q_1 - 2q_2\}$.
- ▶ II wishes to:

$$\min_{q_1, q_2} \max\{q_1 + 6q_2, 4 - q_1 - 2q_2\}$$

- ▶ The textbook shows the a 3D graph of II's solution.
- ▶ We'll solve it algebraically.

II's security strategy

- ▶ $\min_{q_1, q_2} \max\{q_1 + 6q_2, 4 - q_1 - 2q_2\} = ?$
- ▶ The previous theorem states that $\underline{v} = \overline{v} \Leftrightarrow \exists$ saddle point.
- ▶ And in the proof, it is shown when $\underline{v} = \overline{v}$, two players' security strategies form a saddle point.
- ▶ From the Minimax Thm., two players' security strategies form a saddle point, or an NE.
 - ▶ II's equilibrium strategy will make I indifferent between s_1 and s_3 :

$$q_1 + 6q_2 = 4 - q_1 - 2q_2.$$

- ▶ We need 1 more equation to solve q_1 and q_2 .